

Vlad Smirnov

Data Engineering Intern

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Master's student in Data Science at HSE FCS with experience in analytics across accounting outsourcing and online education. Currently focused on data engineering and building projects around ETL pipelines, distributed data processing, backend services, and DevOps and SRE practices. Particularly interested in understanding how data systems work under the hood and how to make them robust and reliable.

PROJECTS

ETL E-commerce Clickstream

Stack: Python, PostgreSQL, ClickHouse, onETL, PySpark, FastAPI, Celery, Redis, Docker, Kubernetes

Built an ETL pipeline for e-commerce data and designed a full path from source systems to an analytical layer. The project loads data into PostgreSQL, reads it with onETL and Spark, processes it in Spark, and writes the result to ClickHouse.

1. Split the pipeline into full and incremental modes, moved both flows into separate Spark jobs, and added HWM tracking in a YAML file. This made it possible to run the first full snapshot separately, rerun the entire dataset, and process only new data after the latest checkpoint.
2. Set up unified service configuration through config.yaml, .env, ConfigMap, and Secret so the same service can run locally and in Kubernetes without manual parameter changes before each launch.
3. Built an HTTP API with FastAPI and background tasks with Celery and Redis. The API makes it possible to trigger full and incremental ETL runs, check the status of a specific task, and keep a history of recent runs.
4. Improved the processing layer, added bot-activity filters, and moved source reading, transformations, and task history into separate services to make the pipeline easier to extend and test.
5. Prepared two launch scenarios, one locally through Docker Compose and one in Kubernetes through Dockerfile, a spark-submit job, and a set of manifests for Minikube.

This project gave me a clearer understanding of how to build an ETL service with two loading modes, keep configuration in one place, and separate local and Kubernetes execution scenarios.

github.com/tadzhnahal/etl-ecommerce-clickstream

Data Vault Posts Loader

Stack: Python, PostgreSQL, Docker Compose, Data Vault 2.0

Built an ELT project that pulls data from a REST API, loads it into the STG layer in PostgreSQL, and moves it into DDS using a Data Vault 2.0 model.

1. Set up local PostgreSQL through Docker Compose, prepared schemas and tables for the STG and DDS layers, and configured the environment for local runs.
2. Wrote two Python scripts: the first loads a fresh snapshot from the REST API into STG, and the second moves rows into DDS and calculates hash keys for hub, link, and satellite tables.
3. Structured the data according to Data Vault 2.0 so that business keys, relationships, and descriptive attributes stay separate, while repeated loads do not duplicate identical post details.

The project helped me better understand how STG and DDS differ, why business keys and relationships are split into separate entities, and what a simple Data Vault 2.0 load looks like.

github.com/tadzhnahal/gazprom-data-vault-posts-task

Flights DWH

Stack: Python, PostgreSQL, Apache Airflow, S3, Jupyter

Built the first part of a flight data warehouse project where the STG layer takes raw files, normalizes fields, and loads data into PostgreSQL.

1. Set up ingestion of flight data from S3 and the airports reference file so that the next warehouse layer works with data already loaded into PostgreSQL instead of raw sources.
2. Built an Airflow DAG for loading and for the overall pipeline run so that execution order and orchestration are handled in one place.
3. Checked the result through SQL and Jupyter to make sure the data lands correctly in STG and is ready for the next warehouse layers.

This project made the role of the first warehouse layer clearer to me, along with the connection between source ingestion, orchestration in Airflow, and preparing the base for the next DWH layers.

github.com/tadzhnahal/flights-dwh

HTTP Status Checker Automation

Stack: Python, Docker, Ansible, Linux

Built a small infrastructure project for HTTP-check automation that includes a Python script, a Docker image, and an execution scenario through Ansible.

1. Wrote a Python script that sends requests to a service with different status codes, logs the responses, and does not stop the whole run if one of the requests fails.
2. Built a Docker image with Python and dependencies so the script can run in a container without manual environment setup on the host.
3. Wrote an Ansible playbook that prepares Docker, builds the image, starts the container, and checks the result through logs.

After this project, it became clearer to me how to build a small infrastructure scenario end to end — from a Python script to a container and an automated run through Ansible.

github.com/tadzhnahal/yadro-infra-platform-task

Reliability Lab

Stack: Python, Bash, Docker, Linux, systemd

Building an educational lab around a small service to study application operations, typical failures, and core SRE scenarios in one place.

1. Added a Docker environment for local startup, moved infrastructure files into separate directories, and prepared a lab that can be launched quickly and reproduced repeatedly.
2. Designed a dedicated systemd failure scenario to investigate service downtime, read logs, find the root cause, and recover the application manually.
3. Organized the project into app, config, infra/docker, labs/linux/systemd-failure, and scripts layers to separate service code, configuration, infrastructure, and scenarios.

The project helps me better understand how to reproduce a service failure, break down a systemd incident step by step, and build a simple educational lab around an infrastructure scenario. I continue to develop the project and gradually add new scenarios.

github.com/tadzhnahal/reliability-lab

EXPERIENCE

Eduson.tv · Analyst-Methodologist

July 2024 — November 2025

Responsibilities: product research, metrics analysis, development and improvement of IT courses.

- Developed 150+ hours of IT content in Python, web development, test automation, and IT project management. Over the year, the cluster expanded its product portfolio by 28% and doubled revenue.
- Ran 15+ filming sessions with experts from Yandex, Sber, Amazon, Booking.com, and EPAM. The materials became part of all cluster programs.
- Reworked 85% of the free Python course content based on A/B tests and conversion analysis. The course generated about USD 6k in revenue within a week after launch.
- Conducted 7+ product studies of new directions and helped launch the development of two courses that exceeded 400% of the monthly revenue target within two weeks after launch.
- Analyzed support processes, identified key bottlenecks, and developed a framework that reduced the average response time to student questions by 60% and cut content development costs by 42%.
- Refined and launched capstone projects for web development, test automation, and strategic IT management courses, one of which became one of the cluster's flagship products by revenue with an average student rating of 4.6/5.

Accounting and Taxes LLC · Junior Data Analyst

January 2024 — July 2024

Responsibilities: data reconciliation, registry preparation, client card checks, and management reporting.

- Built a unified register for 30 clients and connected services, payments, documents, and reporting statuses from 1C, Excel, and EDI. This helped the manager see debts, delays, and at-risk clients faster, while the team reduced data search time by 30%.
- Prepared a package of SQL instructions with join, group by, case when, and subqueries to reconcile payments, services, and reporting periods. The team found unpaid invoices worth about USD 800–1.2k per quarter faster, while overdue receivables in the work area decreased by 12–15%.
- Found duplicates, empty fields, and errors in client cards and registers. Accountants reduced manual reconciliation work by 6 hours per week and handled more client tasks without losing quality.
- Prepared a management report on reporting deadlines and submission statuses. The team started spotting deadline risks earlier, and the number of urgent end-of-month tasks dropped by 20%.

EDUCATION

HSE University Master's degree, Applied Mathematics and Computer Science	2025–2027
Saint Petersburg State University Master's degree, Media Communications	2023–2025
Voronezh State University Bachelor's degree, Journalism	2018–2023

ADDITIONAL EDUCATION

VK Education · Applied Data Analytics Studying Big Data, ETL, Data Quality, BI systems, SQL, Python, scoring, prioritization, and hypothesis work. The course helps me better understand how data moves from a source to a product decision.	2026 — present
VK Education · DevOps/SRE Engineer Studying Linux, networks, web architecture, Docker, Kubernetes, service monitoring, and typical infrastructure failures. The course helps me better understand production infrastructure, diagnose failures more confidently, and work with service reliability more accurately.	2026 — present
VK Education · Distributed Computing Studying network protocols, query optimization, sharding, and replication. Next, I will study NoSQL, Apache Kafka, SRE, observability, and load testing. The course helps me better understand how high-load systems work and how to make them resilient.	2026 — present
Greenatom · Cloud Services Studying VK Cloud services through hands-on work with networks, virtual machines, load balancing, Object Storage S3, Managed PostgreSQL, OpenStack, and basic automation. The program helps me better understand cloud infrastructure, networking scenarios, and how data services work in the cloud.	2026 — present

TOOLS

python

bash

sql

postgresql

clickhouse

excel

google sheets

pandas

numpy

matplotlib

seaborn

pyspark

onetr

apache airflow

fastapi

celery

redis

docker

kubernetes

ansible

terraform

openstack

jupyter

linux

git

LANGUAGES

English	B2
Russian	native

